

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CL)

May 2012

CL 205 Building Materials and Construction

Date: 05.05.2012, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Enlist various properties of engineering materials. Explain any one in detail. [5]
(b) Draw a neat sketch of cross-section of an exogenous tree. [2]
- Q - 2 (a) What are various uses of plastics? Discuss the uses of plastics in the construction field. [5]
(b) Write stepwise procedure for manufacturing of bricks. [5]
(c) Explain important terms used in stairs. [4]

OR

- Q - 2 (a) What are the uses of rubber? What are the requirements of a mortar? [5]
(b) Explain the properties of a good bricks. [5]
(c) What are the requirements of good stair? [4]
- Q - 3 (a) Write a short note on solid ground floor. [5]
(b) Draw a neat sketch of king post truss showing all components. [5]
(c) Draw neat sketch of types of steps. [4]

OR

- Q - 3 (a) Enlist different flooring material. What are factors affecting selection of a flooring material? [5]
(b) Draw a neat sketch of queen post truss showing all components. [5]
(c) Give the detailed classification of stairs. What should be the standard dimension of a stair? [4]

SECTION - II

- Q - 4 (a) Briefly explain various types of pointing with neat figures. [5]
(b) Give detailed classification of foundation. [2]
- Q - 5 (a) What is the need to provide foundation? What are the essential requirements of it? [7]
Discuss the causes of failures of foundation and remedial measures to prevent it?
(b) Draw an elevation and plan for alternate courses of English bond (1 brick thick) wall. [7]

OR

- Q - 5 (a)** Write a short note on pre-cast concrete piles. [3]
- (b)** Draw a neat sketch of combined rectangular R.C.C. footing with longitudinal beam. [4]
- (c)** Write short notes on following types of stone masonry: [7]
- (i) Random (un-coursed) rubble (ii) Coursed rubble (iii) Ashlar chamfered fine tolled
- Q - 6 (a)** Draw neat sketches of following: (**Any Two**) [9]
- (i) Double-Leaf Six Paneled Door
- (ii) Casement Window
- (iii) Fan Light Ventilator
- (b)** Define the terms stretcher, bond, header bond and joint. [2]
- (c)** Write short notes on any **Two**: [3]
- (i) Vertical sheeting
- (ii) Pit method of underpinning
- (iii) Mason's scaffolding
- (iv) Raking shore

OR

- Q - 6 (a)** Draw neat sketch: (Attempt **Any Two**) [9]
- (i) Battened, Ledged and Braced Door
- (ii) Solid Core Flush Door
- (iii) Bay window
- (b)** Differentiate between junctions and quoins in connections for brick bond. [2]
- (c)** Write short notes on any **Two**: [3]
- (i) Runner system
- (ii) Pile method of underpinning
- (iii) Cantilever scaffolding
- (iv) Single flying shore
